



VALIDATION REPORT

ZANO SOLAR POWER



Document Prepared by: (LGAI Technological Center, S.A. (Applus+ Certification))

Project Title	<i>Zano solar power</i>
Version	5.0
Report ID	VCS 4389

Report Title	<i>Zano solar power</i>
Client	<i>AERA Group</i>
Date of Issue	<i>13-September-2024</i>
Prepared By	LGAI Technological Center, S.A. (Applus+ Certification)

Contact	<i>Campus UAB – Ronda de la Font del Carme, s/n</i> <i>08193 Bellaterra – Barcelona (Spain)</i> <i>Tel.:+34 93 567 20 08</i> <i>Fax.:+34 93 567 20 01</i> <i>www.appluscertification.com</i> <i>agustin.calle@applus.com</i> <i>carla.debat@applus.com</i>
Approved by	<i>LGAI Technological Center S.A. (Applus+ Certification) VVB Technical Manager – Mr. Agustín Calle de Miguel</i>
Work carried out by	<i>Pankaj Kumar - Lead Auditor/Technical Expert</i> <i>Technical Reviewer- Srikanth Meesa</i>

Summary:

Validation purpose: The project Zano solar power undertaken by AERA group is renewable energy facility in the country Burkina Faso. Zano is located close to the town of Tenkodogo in the Bougou region of Central-Eastern Burkina Faso. Power plant will be Operated by Quadran Burkina Faso, the project entails the installation of photovoltaic (PV) panels with a total capacity of 24MWc. These panels will harness solar energy, generating electricity for exportation to the West African Power Pool via the national grid. Thus, the project will contribute to the reduction of GHG emissions by displacing fossil fuel-dominated grid-based electricity with renewable energy – solar energy based renewable electricity.

The start date of the project activity is the project's COD (Commercial Operation Date) which is 24-October-2023, which is the date on which the project started generating emission reductions. An undertaking has been submitted by PP for double counting confirming that no GHG reduction will be claimed in any other GHG mechanism. Project activity has undergone continued operation and no major breakdown had taken place except for routine maintenance/shutdown.

The project activity, which has a lifetime of 25 years, is expected to produce 44,659 MWh of electricity annually. The total estimated emission reductions of the project are 179,124 tCO_{2e} during the first crediting period. The ex-ante emission reduction calculation shows that the project will avoid around 25,589 tCO_{2e} annually.

The project will not only reduce GHG emissions but is expected to achieve many social benefits such as improving social services to targeted social groups and generating job opportunities. Many job opportunities have already been created for the local people that are intended to promote sustained, inclusive, and sustainable economic growth. The project also helped ensure that the local stakeholders acquire the knowledge and skills needed to promote sustainable development, lifestyles, human rights, and gender equality through education.

The objective of this validation activity is to have an independent third party for the assessment of the project design, and estimated ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. In particular;

- The project's monitoring plan is assessed against "ACM0002: Grid-connected electricity generation from renewable sources --- Version 21.0.
- The project's compliance with the requirements of VCS guideline version 4.4 and standard version 4.5.
- CDM Validation and Verification Standard for project activities, version 03.0

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified emission reductions (VCUs).

A risk-based approach has been followed to perform this validation activity. In the course of validation, 07 Corrective Action requests (CAR), 01 Clarification Requests (CLs), and 01 Forward action requests (FARs) were raised. The review of the VCS PD and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews, and PP have provided VVB with sufficient evidence to verify the fulfillment of the stated criteria of VCS.

CONTENTS

1	INTRODUCTION	5
1.1	Objective	5
1.2	Scope and Criteria	5
1.3	Reasonableness of Assumptions	6
1.4	Summary Description of the Project	6
2	VALIDATION PROCESS	7
2.1	Method and Criteria	7
2.2	Document Review	9
2.3	Interviews	9
2.4	Site Visits	10
2.5	Resolution of Findings	11
3	VALIDATION FINDINGS	12
3.1	Project Details	12
3.2	Safeguards	16
3.3	Application of Methodology	19
3.4	Non-Permanence Risk Analysis	50
4	VALIDATION OPINION	50
	APPENDIX 1: DOCUMENTS REVIEW UNDER VALIDATION	52
	APPENDIX 2: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)	55
	APPENDIX 3: COMPETENCE OF TEAM MEMBER AND TECHNICAL REVIEWER	61
	APPENDIX 4: ABBREVIATIONS	62
	APPENDIX 5: STAKEHOLDER CONSULTATION	63

1 INTRODUCTION

1.1 Objective

LGAI Technological Center S.A. (Hereinafter referred as Applus + Certification) has been appointed by “AERA group” to perform the validation of the project entitled “Zano Solar Power “under VCS standard V 4.5/^{11/}and VCS guideline version 4.4/^{11/} The objective of this validation activity is to have an independent third party for the assessment of the project design, ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable VCS and CDM (when applicable) requirements. In particular;

The project’s baseline is assessed against “ACM0002: Grid-connected electricity generation from renewable sources — Version 21.0/^{11/}

The projects compliance with the requirements of VCS guideline V4.4/^{11/} and standard version V4.5.^{11/}

CDM Validation and Verification Standard for project activities version 03.0/^{11/}

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated verified certified units (VCUs).

1.2 Scope and Criteria

The scope of the validation is the independent and objective review of the Project Description. The VCS PD is reviewed against the relevant criteria (see 1.1) and decisions by the CDM Executive Board and VCS Executive Board, including the approved baseline and monitoring methodology. The validation was based on the guidance given in the VCS guideline version 4.4/^{11/} and standard version 4.5.^{11/}

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the VCS PD. The main focus of the assessment team is to identify the significant risks for the project implementation and the generation of VCUs. The validation is not meant to provide any consulting towards the Project Proponents. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The only purpose of the validation is its usage during the registration process as part of the VCS project cycle. Therefore, LGAI Technological Center S.A. (Applus+ Certification) can’t be held liable by any party for decisions made or not made based on the validation opinion, which will go beyond that purpose.

1.3 Reasonableness of Assumptions

The validation has been planned and organized to determine the reasonableness of the assumptions, limitations, and methods that support the PP's claims about achievement of the Emission Reductions for the Project.

The Validation Team determined that the claims made for this VCS Project are according to the VCS regulatory requirements and are suitable to occur during the operation of the Project, hence the claims are reasonable, including the assumptions, limitations and methods established by the PP supporting these claims.

1.4 Summary Description of the Project

The project "Zano solar Power" undertaken by AERA group is in West Africa and the first renewable energy facility in the country. The plant is which is located in Zano that is located near the town of Tenkodogo, in Bougou region, Center- Eastern Burkina Faso Burkina Faso and geo-coordinates of the project location are 11°42'13248" N; - 0°22'9444" E. The capacity of the plant is 24 MWc. This plant supplies reliable, clean electricity to the national grid which is then supplied to hundreds of thousands of homes and businesses in the country and saves CO₂ emissions. Thus, the project has contributed to the reduction of GHG emissions by displacing fossil fuel-dominated grid-based electricity with renewable energy –solar energy based renewable electricity.

The start date of the project activity is the project's COD (Commercial Operation Date) which is 24-October-2023, which is the date on which the project started generating emission reductions.

The main elements of the solar power plant are:

- 43 960 Modules: LONGI LR5-72HPH- 545M/550M;
- 396 Trackers: PVH Axone-DUO
- 88 Inverters: SUNGROW250HX
- 3 Transformers station: SUNGROW MVS6750 – 33KV/0.8KV

The solar panels specifications are detailed below:

Make	Longi
Model	LR5-72HPH 530~550M
Power	550 Wp
Max. voltage	49.80 V

Max. intensity	13.98 A
Efficiency	21.5 %
Lifetime	25 years warranty for extra linear power output

Technical specifications mentioned above were crosschecked and confirmed by the assessment team with a technical datasheet^{/6/} submitted by the PP.

The project activity, which has a lifetime of 25 years, is expected to produce 44,659 MWh of electricity annually. The total estimated emission reductions of the project are 179,124 tCO_{2e} during the first crediting period. The ex-ante emission reduction calculation shows that the project will avoid around 25,589 tCO_{2e}, annually.

2 VALIDATION PROCESS

2.1 Method and Criteria

Validation Scope: The scope is defined as an independent and objective review of the project design document. The VCS PD is reviewed against the criteria stated in Article 12 of the Kyoto Protocol, the CDM modalities and procedures as agreed in the Marrakech Accords and the relevant decisions by the CDM Executive Board and VCS standard version 4.5^{/11/} and guideline version 4.4,^{/11/} including the approved baseline and monitoring methodology “ACM0002 - Large-scale Consolidated Methodology, Grid-connected electricity generation from renewable sources (Version 21.0)”^{/11/}. The validation was based on the requirements in the Validation and Verification Standard for project activities version 03.0,^{/11/} and VCS guideline version 4.4^{/11/} and standard version 4.5.^{/11/}

The validation is not meant to provide any consulting towards the Project Proponents. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project document.

Validation Process: The project assessment is based on the Clean Development Mechanism Validation and Verification Standard for project activities version 03.0^{/11/} and VCS standard version 4.5^{/11/} and guideline version 4.4^{/11/} and is conducted using standard auditing techniques to assess the correctness of the information provided by the Project Proponents. Before the assessment begins, members of the team covering the technical scope(s), sectoral scope(s), and relevant host country experience for evaluating the VCS project activity are appointed.

- Once the project is received by the assessment team, the members of the assessment team carried out:
- A desk review of the Project Design Document;
- Follow-up interviews with project stakeholders;

- The resolution of outstanding issues and the issuance of the final Validation report and opinion.

In order to ensure transparency, assumptions must be clear and stated explicitly and background material must also be referenced. LGAI TECHNOLOGICAL CENTER S.A. (APPLUS+ CERTIFICATION) has developed a specific checklist customized for the project. The checklist demonstrates, in a transparent manner, the project criteria (requirements), discussion on each criterion by the assessment team, and the results from validating the identified criteria.

Appointment of the assessment team:

According to the sectoral scope / technical area and experience in the sectoral or national business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification).

The composition of audit team shall be approved by the LGAI Technological Center S.A. (Applus+ Certification) ensuring that the required skills are covered by the team.

The qualification levels for Assessment Team members that are assigned by aforementioned appointment rules are as presented below:

- Lead Auditor (LA).
- Technical Expert (TE).
- Financial Expert (FE)
- Technical Reviewer (TR).

The Sectoral Scope / Technical Area required knowledge linked to the applied methodology(ies) is covered by the Assessment Team as shown below:

Name	Role	SS Coverage	TA Coverage	Financial aspect	Host country experience
Mr. Pankaj Kumar	LA/TE/FE	YES	YES	YES	YES
Mr. Srikanth Meesa	TR	YES	YES	YES	NA

The complete list of CVs is included as Appendix 3 of this report.

Document Review:

The VCS PD^{7/} submitted by the Client was reviewed against the approved methodology and other relevant criteria to verify the correctness, credibility, and interpretation of the presented information. Furthermore, a cross-check between information provided and information from

other sources has been done. A complete list of all documents and evidence material reviewed is included in this report below in appendix 1.

Follow -up Interviews:

A remote audit is conducted by LGAI Technological Center S.A. (Applus+ Certification) assessment team who performed interviews, telephone conferences, and remote inspection through ICT means with project stakeholders to confirm selected information and to resolve issues identified in the document review. The detail is provided in this report in the below sections.

Resolution of Clarification and Correction Action Request:

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification) positive conclusion on the project design. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the Client and Applus+ Certifications to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in the appendix 2.

The VCS PD Version 3.1 dated 06-June-2024 submitted by Project Proponents serves as the basis for the final assessment presented. Additional changes to the project during the validation process is not considered to be significant with respect to the main VCS objectives. The two VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Internal quality Control:

As final step of a validation, the final documentation including the final validation and the checklist may undergo an internal quality control by the technical review committee, i.e., each report has to be finally approved either by the head of the technical review committee or the deputy. In case one of these two persons is part of the assessment team approval can only be given by the other one to avoid any conflict of Interest.

After confirmation of the project owners, the positive validation opinion and relevant documents are submitted to the project proponent or its representatives for the submission to the VCS secretariat through the VCS web platform.

2.2 Document Review

As a part of audit, the assessment team conducted validation to confirm the project description based on the supporting documents submitted and to assess whether they are as per the state requirements. The details of the document observed during the validation process are listed below in appendix 1 of this report.

2.3 Interviews

As a part of audit, the assessment team conducted interview of the following stakeholders. The identified personnel, their roles who were interviewed during remote audit who provided information additional to that provided in the project description and any supporting documents is given below;

Date of Interview: 26-June-2023			
Sr. No.	Name of Persons	Role/Designation	Subject
1.	KERE Barké Abdoulaye	Doyen/Représentant coutumier	Project Implementation, Local stakeholder consultation, data storage, QA/QC Procedure, Additionality Project operation, its impact on the local area, project's impact on sustainable development, any grievances.
2.	YAMNEDA Alassane	Président CVD (Conseil Villageois de Développement)	
3.	KERE Adama	Ancien Conseiller municipal du village	
4.	KERE Souleymane	Chef du village	
5.	YAMNEDA Kassoum	Imam du village	
6.	KERE Habibou	Représentante des femmes	
7.	Alexandra Dunod	AERA Carbon	
8.	Gaïa LUDINGTON	AERA Carbon	

2.4 [Site Visits](#)

For the project activity, an exemption from site visit was granted by VERRA via communication letter/14/ dated 31-July-2023. Hence, VVB has conducted a remote audit on 26-June- 2023.

The method and objectives of remote audit, details of all project activity locations visited, the physical and organizational aspects of the project inspected and the dates when such site inspections took place is mentioned in the below table.

Remote Audit (Video conferencing): 26-June-2023				
No.	Activity performed	Site location	Date	Team member
1.	Assessment team checked the implementation of the project, Baseline emission, Emission reduction calculation, Additionality, technical description of the project and Monitoring. Assessment team also checked that whether the monitoring plan as described in the VCS PD is actually practised onsite. Also, assessment team checked any change in host country criteria	Zano (Burkina Faso)	26-June-2023	Mr. Pankaj Kumar

	which may affect the baseline of the project activity.			
--	--	--	--	--

2.5 Resolution of Findings

The objective of this validation is to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification)'s positive conclusion on the project design and emission reduction report. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the Client and LGAI Technological Center S.A. (Applus+ Certification) to guarantee the transparency of the validation process, the concerns raised and responses given are summarized below in Appendix 2.

The final VCS PD Version 03.1 and ER sheet version 03 submitted by project owners on 06-June-2024 serves as the basis for the final assessment presented. Additional changes to the project during the validation process are not considered to be significant with respect to the main VCS objectives. The two VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation findings	No. of CL	No. of CAR	No. of FAR
Project design document	00	01	01
Description of project activity	00	01	00
Application of selected baseline and monitoring methodology and selected standardized baseline			
- Applicability of methodology and standardized baseline	00	01	00
- Deviation from methodology	00	00	00
- Clarification on the applicability of methodology, tool, and/or standardized baseline	00	00	00
- Demonstration of additionality	01	01	00
- Emission reductions	00	01	00
- Monitoring plan	00	01	00
- Stakeholder's consultation process	00	01	00
- Public comments	00	00	00
Others (please specify)-Matter related to double counting- for validation	00	00	00
Others (please specify)- SDG Goals	00	00	00
Total (Validation)	01	07	01

The list of findings and their resolution is presented in Appendix 2 of this report.

2.5.1 Forward Action Requests

This is validation of the project activity and 01 FAR was raised during this validation process. Please refer Appendix 2 for details.

3 VALIDATION FINDINGS

3.1 Project Details

Zano Solar Power Plant is the construction and operation of greenfield solar power plant of 24MWc. This is crosschecked and confirmed by the assessment team with the <https://www.power-technology.com/data-insights/power-plant-profile-zano-solar-pv-park-burkina-faso/?cf-view>^{10/} Zano, positioned near the town of Tenkodogo in the Bougou region of Center-Eastern Burkina Faso, is designated for operation by Quadran Burkina Faso. The project encompasses the establishment of photovoltaic (PV) panels with a total installed capacity of 24MWc, facilitating the capture of solar energy and electricity production for exportation to the West African Power Pool through the national grid.

The baseline scenario is the electricity generated and delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources.

The project activity, which has a lifetime of 25 years^{6/}, is expected to produce 44,659 MWh^{9/} of electricity annually. The total estimated emission reductions of the project are 179,124 tCO_{2e} during the first crediting period of 7 years from 24-October-2023 to 23- October-2030. The ex-ante emission reduction calculation shows that the project will avoid around 25,589 tCO_{2e} annually.

The coordinates of the project location crosschecked by the assessment team via google maps and file shared by the PP and are 11°42'13248"N, 0°22'9444" E. The energy generated by this project, is supplied to the national grid through western African power pool.

According to the latest VCS Standard, version 4.5^{11/}, the project is eligible under the scope of the VCS Program because:

- It results in CO₂ emission reductions, one of the seven greenhouse gases under the Kyoto Protocol
- It consists of grid-connected electricity generation using (large scale) solar power plant built in a Least Developed Country.

The project has been designed to include a single installation of an activity and it includes a single location or installation only.

The validation team has checked and found that the proposed project is not a grouped project.

The validation team checked remotely and confirms that the details of the project proponent areas below:

Organization name	AERA Group
Contact person	Alexandre Dunod
Title	COO
Address	28 cours Albert 1er, 75008 Paris, France
Telephone	+33 1 42 18 02 02
Email	a.dunod@aera-group.fr

Organization name	QAIR (previously Quadran International)
Role in the project	Sponsor
Contact person	Abdoulaye TOURE
Title	Project Director
Address	155 Bd Haussmann 75 008 Paris FRANCE
Telephone	+ 33 1 79 35 67 14 / + 33 6 98 77 99 87
Email	a.toure@qair.energy

Organization name	Quadran Burkina Faso
Role in the project	SPV for Zano project
Contact person	Bruno BARITELLO
Title	Managing Director / Country Director
Address	Projet ZACA Ouagadougou, BURKINA FASO
Telephone	+226 65 89 92 75 / 25 33 33 77

Email

b.baritello@qair.energy

Project Start Date:

The project's COD (Commercial Operation Date) is 24-October-2023, which is the date on which the project started generating emission reductions. The validation team checked the commercial operation date (CoD) from the statistique production report which is submitted by Project Proponent.

Project Crediting Period:

The validation team confirms that the crediting period dates for the project is as below:

The project's crediting period is 7 years renewable.

The first crediting period starts from 24-October-2023 and lasts till 23-October-2030

Project scale & estimated GHG Emissions Reductions or Removals:

The validation team confirms that the project involves setting up single installation of a project activity project with large scale SPVs. The installed capacity of the project is 24 MWc. The estimated annual GHG emission reductions/removals of the project are under 20,000 – 100,000 tCO_{2e}/year.

As per the section 3.9.1 of VCS standard version 4.3/¹¹/, the projects are classified as follows:

1. Projects: Less than or equal to 300,000 tonnes of CO_{2e} per year
2. Large Projects: Greater than 300,000 tonnes of CO_{2e} per year

Hence, the proposed activity instances are classified as “Project”.

Project Scale

Project

✓

Large project

Year	Estimated GHG emission reductions or removals (tCO _{2e})
Year 2023 (From 24-October-2023 to 31-December-2023)	4,936

Year 2024 (1-January-2024 to 31-December-2024)	25,967
Year 2025 (1-January-2025 to 31-December-2025)	25,825
Year 2026 (1-January-2026 to 31-December-2026)	25,682
Year 2027 (1-January-2027 to 31-December-2027)	25,541
Year 2028 (1-January-2028 to 31-December-2028)	25,401
Year 2029 (1-January-2029 to 31-December-2029)	25,261
Year 2030 (From 1-January-2030 to 23-October-2030)	20,511
Total estimated ERs	179,124
Total number of crediting years	7
Average annual ERs	25,589

Conditions Prior to Project Initiation:

The validation team during the desk review and remote assessment confirms that in accordance with methodology ACM0002, Version 21 the project activity is the installation of a new grid-connected renewable power plant (Greenfield). The baseline scenario is defined as electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system” version 7.0

Project compliance with applicable laws, statutes and other regulatory framework:

The validation team confirms that as of the present date, there exist no regulatory constraints or policies impeding the commencement of the project activity. Burkina Faso has prominently underscored the advancement of its renewable energy sector, capitalizing on its substantial solar and wind energy reserves. The inauguration of the National Agency for Renewable Energies and Energy Efficiency in December 2016 marked a pivotal stride in the government's advocacy for

renewable energy initiatives¹. A primary aim of this agency is to diminish the nation's reliance on fossil fuel-based energy sources

Project Ownership:

As per Power Purchase Agreement/^{2/} dated 20 November 2019 the legal right to control and operate the project activity of Zano solar power is with the Quadran Burkina Faso. Validation team has checked the PPA and it was found acceptable.

Participation Under Other GHG Programs:

The validation team found the declaration/^{12/} regarding the Net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits submitted by Project Proponent correct and confirms that, the project activity is not involved in other Emissions trading programs and other binding limits.

Leakage Management for AFOLU project:

Not applicable to the project activity.

Commercially Sensitive Information:

No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the validation team for analysis which lead to positive conclusion for this validation.

Sustainable Development:

The project aligns with host country Burkina Faso's sustainable development goals by improving energy security, enhancing health by reducing fossil fuel emissions, creating local employment opportunities, and facilitating technology transfer by implementing solar energy power plant.

3.2 Safeguards

3.2.1 No Net Harm

The activities during construction, operation, and decommissioning of the project have been analysed to identify corresponding environmental and social impacts and assess their significance. Mitigation measures have been identified in the Environmental and Social

¹ <https://www.usaid.gov/powerafrica/burkina-faso#:~:text=In%202016%2C%20the%20GOBF%20created,the%20use%20of%20renewable%20energy.>

Management Plan (ESMP) for negative impacts where institutional responsibilities for implementation and follow-ups have been presented. The socioeconomic benefits of the project, therefore, outweigh the potential negative environmental and social impacts.

3.2.2 Local Stakeholder Consultation

As per the VCS requirements, it is necessary to invite the relevant stakeholders, prior of the validation process. The stakeholder meeting took place at Zano on 21-February-2020.

The objective of the Stakeholders' Consultation event was to take due diligence of all comments, questions, and opinions in an open and transparent manner towards the projects' economic, social, and environmental impacts, and sustainability. It was also to raise the stakeholders' awareness level about the procedure to register such projects under the VCS.

The Local Stakeholder Consultation report is devised to provide the outcomes of the stakeholder consultation and to ensure the transparency of the process through documentation.

The invitees were sent the invitation to attend the stakeholders' consultation event. They were also requested to submit their written comments, should they have any,

The validation team verified the documents submitted by the PP like the attendance sheet/^{13/} invitation letter/^{13/}, and feedback/^{13/} provided by the local stakeholder and the Project aims to resolve all concerns within 15 days. Thus, the validation team is of opinion that the local stakeholder consultation is adequate and appropriate.

The ongoing communication mechanism, as described in the VCS PD Section 2.2., has been included as a topic on the LSC process and this has been found acceptable.

3.2.3 Environmental Impact

The project activity is having positive impacts and no significant adverse environmental impacts are foreseen. Since, the project activity is an electricity generation from renewable source (i.e., solar energy) therefore no negative impacts are envisaged.

An Environmental and Social Impact Assessment (ESIA) was carried out for the project. The activities during the construction and operation of the project have been analyzed to identify correspondent environmental impacts and assess their significance.

After careful examination of the project activities and the surrounding environment, it is determined that no major impacts are anticipated. The nature of the receptors involved suggests that most of the potential impacts, such as those related to air quality, noise emission, water resources, soil, biodiversity, landscape, land use, and access, are either negligible or minor. Mitigation measures have been applied effectively, further reducing the expected magnitude of these impacts. Therefore, management should not anticipate anything other than the implementation of sectorial best practices to manage these impacts. A summary of the expected magnitude of impacts per category, both before and after mitigation, is provided in the table below:

Subject to impact	Impact mitigation before	Residual impact
Air quality	Minor (construction)	Negligible
	Negligible (operation)	
	Positive (Greenhouse gas)	Positive
Noise	Moderate (construction)	Minor
	Minor (operation)	Negligible
Water resources (consumption)	Minor	Negligible
Water resources (drainage and erosion)	Moderate (construction)	Minor
	Negligible (operation)	Negligible
Soil	Minor (construction)	Negligible
	Negligible (operation)	
Biodiversity flora	Moderate (construction)	Minor
	Negligible (operation)	Negligible
Biodiversity fauna	Minor	Negligible
Landscape	Minor (construction)	Minor
	Minor (operation)	
Land use	Minor	Minor
Roads and access	Negligible (construction and operation)	Negligible
Agriculture, livelihood, local economy	Moderate	Minor

Community Health and Safety	Moderate (construction)	Minor
	Minor (operation)	Negligible
Workers Health and Safety	Moderate (construction)	Minor
	Negligible (operation)	Negligible
Waste management	Minor (construction)	Negligible
	Minor (operation)	
Traffic	Minor (construction)	Negligible

3.2.4 Public Comments

The validation team noted that this project was open for public comment from 02-June-2023 to 02-July-2023. No comments were received.

The details were checked by the validation team in the following web platform.²

3.2.5 AFOLU-Specific Safeguards

Since it is a non-AFOLU project, this section is not required.

3.3 Application of Methodology

3.3.1 Title and Reference

Assessment team checked that following methodology and tools are applicable for the project activity. The details are as below:

Methodology: ACM0002: Grid-connected electricity generation from renewable sources — Version 21/11/

Project Type: Renewable energy projects (Large Scale)

Sectoral Scope: 1

The methodology refers to the following methodological tools that have been considered by the project):

² <https://registry.verra.org/app/projectDetail/VCS/4389>

TOOL01: Tool for the demonstration and assessment of additionality, version 07.0/^{11/}

TOOL 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”, version 3.0/^{11/}

TOOL07: Tool to calculate the emission factor for an electricity system, version 07.0/^{11/}

TOOL 24: Common practice analysis, version 3.1/^{11/}

TOOL 27: Investment analysis, version 12.0/^{11/}

3.3.2 Applicability

The applied baseline methodology is justified as it has been demonstrated that the proposed project activity is:

For ACM0002: Grid-connected electricity generation from renewable sources --- Version 21/¹¹

Methodological Applicability	Project Justification	VB Assessment
This methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Install a Greenfield power plant; (b) Involve a capacity addition to (an) existing plant(s); (c) Involve a retrofit of (an) existing operating plants/units; (d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or (e) Involve a replacement of (an) existing plant(s)/unit(s).	The project activity is a new grid connected renewable energy power plant (Solar) at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield plant) and hence this criterion is applicable.	The project activity is a new grid connected renewable energy power plant (Solar) at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield plant) and hence this criterion is applicable.
In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant;	The project activity does not involve the integration of BESS.	The project activity does not involve setting up and implementation of a BESS, which was confirmed through remote inspection with the PO representatives.

(b) Integrate a BESS together with implementing a capacity addition to (an) existing solar photovoltaic or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); (d) Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).		Hence this criterion is not applicable.
The methodology is applicable under the following conditions: a) The project activity may include renewable energy power plant/unit of one of the following types: hydropower plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity. c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the Project Proponents shall	Option (a) – The project activity is the construction and operation of two solar power plants hence the methodology is applicable. Option (b) - The project activity does not involve any capacity additions, retrofits, rehabilitations or replacements. Option (c)- The project does not involve BESS. Option (d) - The project does not involve BESS.	The project activity a new grid connected renewable energy power plant (Solar) at a site where no renewable power plant was operated prior to the implementation of the project activity (Greenfield plant) and hence this criterion is applicable. Validation team checked O & M contract confirm that the plant is a greenfield renewable energy power plant. The project activity does not involve any capacity additions, retrofits or replacements and therefore this

demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); d) The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies 2 may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section 5.4.4 below. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the Project Proponent shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period.		second condition is not applicable.
---	--	--

In case of hydro power plants, one of the following conditions shall apply: a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density, calculated using equation (7), is greater than 4 W/m²; or c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7), is greater than 4W/m²; or d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7), is lower than or equal to 4W/m², all of the following conditions shall apply: i. The power density calculated using the total installed capacity of the integrated project, as per equation (8), is greater than 4 W/m²; ii. Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; iii. Installed capacity of the power plant(s) with power density lower than or equal to 4W/m² shall be a) Lower than or equal to 15 MW; and b) Less than 10 per cent of the total installed capacity of integrated hydro power project.	Not applicable, the project is not an integrated hydro power project as it concerns solar power plants.	The project activity is a grid connected renewable energy project (Solar). This condition is applicable only for hydro power plants and not applicable for solar projects. Hence this criterion is not applicable.
---	--	---

In the case of integrated hydro power projects, project proponent shall: a) Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or b) Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior to the implementation of the CDM project activity.	Not applicable, the project is not an integrated hydro power project as it concerns solar power plants.	The project activity is a grid connected renewable energy project (Solar). This condition is applicable only for hydro power plants and not applicable for solar power projects. Hence this criterion is not applicable.
The methodology is not applicable to: a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; b) Biomass fired power plants/units.	The proposed project activity neither involves: - switching from fossil fuels to renewable energy sources at the site of the project	The project activity is installation of a new grid connected renewable energy project (Solar) and does not involve switching from fossil fuel to renewable energy and hence this

	activity, since in this case the baseline may be the continued use of fossil fuels at the site, nor - biomass fired power plants/units.	criterion is not relevant. The project activity does not involve any biomass-based power plants and hence this criterion is not applicable to the project activity.
In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	The project activity does not involve capacity additions, retrofits, rehabilitations or replacements.	The project activity a new grid connected renewable energy plant (Solar) and not a retrofits, replacement or capacity additions and therefore this criterion is not applicable to the project activity.

Compliance of the project activity regarding applicability conditions of Standardized Baseline ASB0034-2021

No	Applicability condition	Applicability to this project activity
(a)	The project activity is implemented in any of the countries members of WAPP and is connected to the project electricity system;	The project activity is implemented in Burkina Faso, member of WAPP, and connected to the same electricity system.
(b)	The CDM approved methodology that is applied to the project activity requires the determination of CO ₂ emission factor(s) through the application of the grid tool;	As part of ACM0002, “operating margin” (OM), “build margin” (BM) and “combined margin” (CM) need to be estimated to calculate baseline emissions of the project activity.

No	Applicability condition	Applicability to this project activity
(c)	The project activity uses ex ante option for the grid emission factor as indicated in the tool i.e. no monitoring and recalculation of the emissions factor during the crediting period is required	Ex-ante option for grid emission factor is applied in this project activity.

TOOL01: Tool for the demonstration and assessment of additionality --- Version 7.0.0 is applicable under the following conditions:

Tool Applicability	Project Justification
The use of the “Tool for the demonstration and assessment of additionality” is not mandatory for Project Proponents when proposing new methodologies. Project Proponents may propose alternative methods to demonstrate additionality for consideration by the Executive Board. They may also submit revisions to approved methodologies using the additionality tool.	Not applicable. No new methodology is proposed.
Once the additionally tool is included in an approved methodology, its application by Project Proponents using this methodology is mandatory.	The additionality tool is applied, hence applicable.

TOOL7 “Tool to calculate the emission factor for an electricity system” version 07.0 is applicable under the following conditions:

Tool Applicability	Project Justification
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g.,	The project is a grid-connected solar energy power generation project and thus substitutes grid electricity.

demand-side energy efficiency projects).	
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the Project Proponents, i.e., option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	steps involved in calculation of Emission Factor is included in section 3.1 of the VCS PD as per the required tool.
In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	The Project is located in non-Annex I country and hence the tool is applicable.
Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	Not applicable. No biofuel is involved in this project activity.

TOOL24 “Common practice” Version 3.1 is applicable under the following conditions:

Tool applicability	Project Justification
This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool	PP has applied “Tool for the demonstration and assessment of additionality”. Hence, the current tool is applicable.

“Combined tool to identify the baseline scenario and demonstrate additionality”, or baseline and monitoring methodologies that use the common practice test for the demonstration of additionality.	
In case the applied approved baseline and monitoring methodology defines approaches for the conduction of the common practice test that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Not applicable. No approach that are different from those described in the tool has been used.

TOOL27 “Investment analysis” Version 12 is applicable under the following conditions:

Tool applicability	Project Justification
This methodological tool is applicable to project activities that apply the methodological tool “Tool for the demonstration and assessment of additionality”, the methodological tool “Combined tool to identify the baseline scenario and demonstrate additionality”, the guidelines “Non-binding best practice examples to demonstrate additionality for SSC project activities”, or baseline and monitoring methodologies that use the investment analysis for the demonstration of additionality and/or the identification of the baseline scenario.	“Tool for the demonstration and assessment of additionality” has been applied. Hence, the tool is applicable.
In case the applied approved baseline and monitoring methodology contains requirements for the investment analysis that are different from those described in this methodological tool, the requirements contained in the methodology shall prevail.	Not applicable. No approach that are different from those described in the tool has been used.

LGAI Technological Center S.A. (Applus+ Certification) confirms that the application of the baseline methodology is transparent and conservative and confirms that the chosen baseline and monitoring methodology is applicable to the project activity.

The project activity qualifies as Type I during every year of the crediting period in accordance with applicable provisions for project activity eligibility as discussed above. Also, the total installed capacity of project activity is 24 MW which is applicable as per large-scale project activities methodologies.

3.3.3 Project Boundary

As per methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 21^{11/}

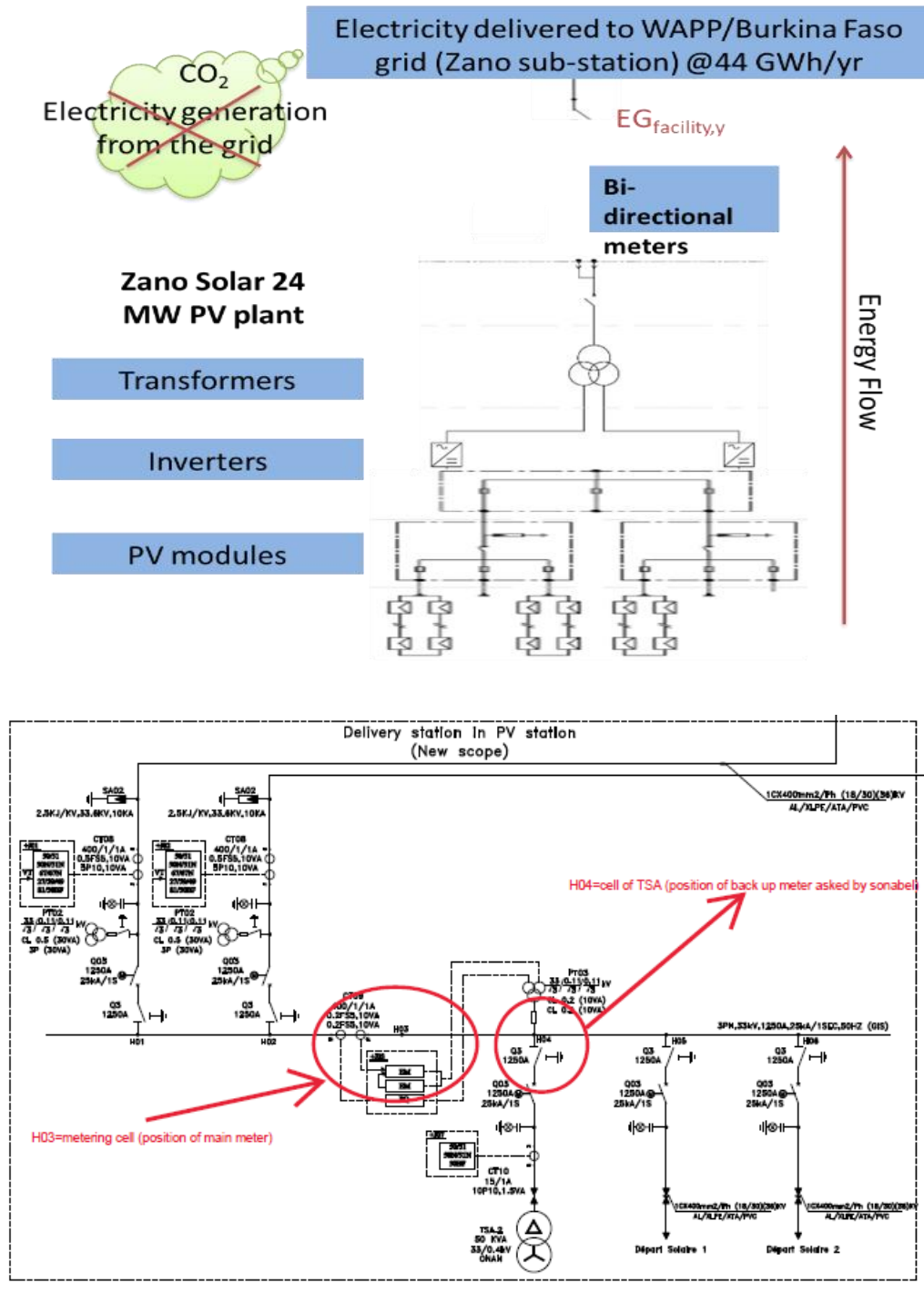
“The spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the solar power plant is connected to”.

The electricity system is defined by the spatial extent of the power plants that are physically connected through transmission and distribution lines to the solar power plant and that are covered by either single or layered dispatch area.

Thus, the project boundary for this activity is the national/sub-national grid in Burkina Faso

The sources and GHG gases involved for the Project activity are as below:

Source	Gas	Included?	Justification/Explanation
Baseline	CO ₂	Included	Main emission source
	CH ₄	Excluded	Minor emission source
	N ₂ O	Excluded	Minor emission source
Project	CO ₂	Excluded	Minor emission source
	CH ₄	Excluded	Minor emission source
	N ₂ O	Excluded	Minor emission source



SLD with meters position

3.3.4 Baseline Scenario

According to the ACM0002: Grid-connected electricity generation from renewable sources --- Version 21^{11/}

“If the project activity is the installation of a Greenfield power plant, the baseline scenario is the electricity delivered to the grid by the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system”, version 07.0.^{11/}

The project activity is the installation of a new power plant at the site and is not a capacity addition or retrofit or replacement of any other existing plant. Any electricity generated and exported from the project activity would in its absence have been generated in the grid dominated by fossil fuel-based power plants, which represents the baseline scenario. The project scenario is a Solar photovoltaic power plant that would generate energy and supply it to the SONABEL electricity grid.

The baseline, the scenario existing prior to the implementation of the project activity, is the continuation of the current practice. The electricity generated by the project activity would have been generated by the addition of new fossil fuel-based generation sources to the grid to supply the additional electricity capacity required.

The validation team has accessed and confirm that PP has identified baseline scenario in line with Tool07 and the assessment on the same is presented below in section 3.3.6. It was confirmed that Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool is applicable for the current project and is correctly applied.

The validation team conclude that, the above-mentioned baseline scenario is in line with the applied methodology “ACM0002: Grid-connected electricity generation from renewable sources - -- Version 21”. Same is also reflected in final PD version 06. Thus accepted.

3.3.5 Additionality

The project activity generates renewable electricity by means of solar PV technology, which is a renewable, zero emission source of energy. Additionality of the project activity is demonstrated through a project specific additionality test. The project is not mandated by any law, statute, or other regulatory framework and the Project Proponent has stated the same in Section 3.5. of the VCS PD^{7/}, which has been confirmed by the assessment team.

Baseline considerations for the project are based on approved consolidated baseline methodology ACM0002 (Version 21.0). The methodology requires the project proponent to determine the additionality based on the “CDM - Methodological Tool “Tool for the demonstration and assessment of additionality”, version 07.0. The tool provides for a stepwise approach to demonstrate and assess the additionality of a project. These steps are:

(a) Step 0 Demonstration whether the proposed project activity is the first-of-its-kind;

(b) Step 1 Identification of alternatives to the project activity;

(c) Step 2 Investment analysis; and

(e) Step 4 Common practice analysis.

The stepwise approach to establish additionality of the project activity has been followed, details of which are provided in the following paragraphs:

Step 0: Demonstration whether the project activity is the first-of-its-kind

According to the CDM – Methodological Tool “Tool for the demonstration and assessment of additionality”, version 07.0, this step is optional. Therefore, no analysis or evidence will be provided under this step. The project proponent will prove the additionality starting from Step 1 below.

Step 1: Identification of alternatives to the project activity consistent with current laws and regulations

Sub-step 1a: Define alternatives to the project activity- Identify realistic and credible alternative(s) available to the Project Proponents or similar project developers that provide outputs or services comparable with the proposed CDM project activity.

The project activity is generating electrical energy using renewable energy resources that are fed into the national electricity grid. Hence, the following alternatives are considered:

Alternative 1: The project activity undertaken without being registered under VERRA.

The option of project activity being undertaken without being registered under VERRA is in compliance with all applicable legal and regulatory requirements and can be a part of the baseline. However, this is not a plausible option as deployment of the project activity faces significant financial barriers established in the subsequent section. The electricity generated from the project activity would have been fed to the Togo national grid, which has existing and new fossil fuel-based power plants that would have been used in the absence of the project activity.

Alternative 2: Continuation of the current situation (No project activity and the equivalent amount of energy would have been produced by the grid electricity system through its currently running power plants and by new capacity addition to the grid)

In such case the project proponent would have withdrawn from an investment in the project activity and continued with its usual business activities. In such case the grid would have continued with the fossil fuel-based power plants resulting in GHG emissions. Hence, existing power plants and new capacity add-ons from fossil fuel-based power plants can be considered as an appropriate, realistic & credible baseline alternative for the project activity.

Outcome of Sub-step 1.a: Identified realistic and credible alternative scenario(s) to the project activity.

Of the two alternatives outlined above, the first alternative is not realistic as project activity as it is not viable without the benefits of carbon credits. The second alternative on the other hand is the baseline scenario for the project activity as per methodology.

Therefore, the continuation of the current situation is the most plausible alternative to the project activity. The project being a green field project activity and the baseline scenario, in line with the methodology, is “the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources

Sub-step 1b: The alternative(s) shall be in compliance with all mandatory applicable legal and regulatory requirements, even if these laws and regulations have objectives other than GHG reductions, e.g., to mitigate local air pollution.

The implementation of the project activity is consistent with mandatory laws and regulations in Burkina Faso and relevant clearances and approvals have been obtained.

Outcome of Sub-step 1.b: Identified realistic and credible alternative scenario(s) to the project activity that are in compliance with mandatory legislation and regulations taking into account the enforcement in the region or country decisions on national and/or sectoral policies and regulations.

Both the alternatives identified above are found to be in compliance with the mandatory laws and regulations taking into account the enforcement of the legislations in the region/country and relevant decisions on national and/or sectoral policies and regulations.

However, Alternative 2 “Continuation of the current situation” has been selected as the appropriate baseline alternative for this project activity in line with the methodology.

Step 2: Investment analysis

An investment analysis is carried out to determine whether the project activity is economically or financially less attractive than at least one other alternative, identified in step 1, without the revenue from the sale of carbon credits. This is demonstrated in line with the tools for the relevant sections as per “Investment Analysis” (version 12.0).

Sub-step 2a: Determine appropriate analysis method

The project activity envisages the export of renewable energy to the SONABEL electricity grid and the revenues from the sale of electricity are based on the terms and conditions within the Power Purchase Agreement (PPA). /2/

A simple cost analysis (Option I) cannot be used as the analysis method as the sale of electricity will result in a revenue stream during the project operation. In the absence of the project activity, the continued usage of the grid electricity would have been the most plausible option, which does not require an investment.

Hence an investment comparison analysis (Option II) is also not appropriate for the project activity.

After eliminating Option, I and Option II, the use of a Benchmark analysis (Option III) is the method of analysis that has been selected as the most suitable method. This method determines the attractiveness of the project activity for the investors, as well as provides a measure of the viability of the investment to generate revenues during its operation, as compared with other avenues and investment options. Hence, the Benchmark analysis method is to be applied in context of the project activity.

Sub-step 2b (Option III): Apply benchmark analysis

The investment analysis using the Benchmark analysis approach (Option III) has been chosen.

Further, this method illustrates that the assessment of the project by the project proponent was carried out prior to the decision to undertake the project was taken and management approval had been granted.

Choice of Financial Indicator:

The project Internal Rate of Return (post-tax Project IRR) is identified as the most suitable financial/economic indicator for the project type and decision context. This indicator allows for effective comparison of the project returns with an appropriate benchmark. Therefore, parameters that (a) are standard in the market and (b) consider the specific characteristics of the project type, but are not linked to the subjective profitability expectation or risk profile of a particular project developer, are the basis for the financial analysis. The benchmark represents the minimum rate of return that would justify the financial viability of the project and therefore its implementation.

According to Methodological Tool27: Investment Analysis version 12.0, all input values were known before the investment decision and can therefore be considered realistic and appropriate values to be used in the financial calculation of the project activity.

Sub-step 2c: Calculation and comparison of financial indicators (only applicable to Options II and III):

the comparison of the real IRR with a real benchmark in the IRR calculation ensures the exclusion of the effects of general price increases due to inflation. Below, an appropriate project benchmark is established using the Weighted Average Cost of Capital (WACC) method, relying on market-standard parameters, given the potential for the Greenfield project activity to be undertaken by other promoters. The Weighted Average Cost of Capital (WACC) is calculated as follows:

$$WACC = re \times We + rd \times Wd \times (1 - Tc)$$

Where:

re = Cost of equity (-)

W_e = Percentage of financing that is equity (-)

r_d = Cost of debt (-)

W_d = Percentage of financing that is debt (-)

T_c = Corporate tax rate (-)

In the absence of a readily available typical debt/equity finance structure observed in the sector of the country, a default assumption of 50% debt and 50% equity financing is made, leading to $w_d = w_e = 50\%$.

The cost of debt is assumed to be the commercial lending rate in the country, as the benchmark relies on market-standard parameters and no documented cost of debt financing of comparable projects is available. Consequently, the latest updated (07-october-2019) average commercial lending-debit rate from West African States' Central Bank statistics, reflecting $r_d = 9.5\%$,³ is chosen.

Regarding the cost of equity, it is determined among the default values for the expected return on equity provided in the Methodological Tool for Investment Analysis version 12.0/^{11/} dated 2 November 2022 (Burkina Faso – Group 1), resulting in $r_e = 17.18\%$.

The applicable corporate tax rate⁴ is determined based on the project financing assumptions, set at $T = 0\%$ for years 1 to 7,

$T = 15\%$ for years 8 to 15

, and $T = 27.5\%$ for years 16 to 25.

Upon calculation, the Weighted Average Cost of Capital (WACC) is determined to be 12.54%, which is higher than the Project Internal Rate of Return (Post tax IRR) of 7.05% in the absence of carbon revenues.

Sr. No.	Parameter	VVB Assessment
1.	Installed Capacity	Installed capacity of the project activity 'Zano solar power' is 24 MW which is assessed from the statuesque production report/ ^{1/} and found acceptable. The project capacity has been cross checked from the yield assessment study/ ^{9/} . The project capacity was also cross verified from the https://www.power-technology.com/data-insights/power-plant-profile-zano-solar-pv-park-burkina-faso/ . Additionally, a FAR is also raised for commissioning certificate.

³ <https://www.bceao.int/sites/default/files/2020-03/CONDITIONS%20DE%20BANQUE%20DECEMBRE%202019.pdf> consulted July 2020

⁴ 25-y average applicable tax rate

2.	Installation cost	Installation cost of the project activity assumed as 24.037 million EUR that was cross verified by submitted facility agreement ^{4/} dated 23-August-2021 and found acceptable to the assessment team.
3.	Investment horizon	According to the submitted technical specifications from manufacture Investment horizon (average life time) of the project activity is expected 25 years. Average life time of the project activity was also cross verified by the yield assessment study which is found consistent and acceptable to the assessment team.
4.	Income tax	PP has considered 16 % income tax rate which is according to the 25-year average applicable tax rate which was checked and found in line with the host country guidelines.

The validation team concludes that the above-mentioned Project IRR mentioned under IRR spreadsheet against the WACC in the investment analysis sheet and assumptions considered for the calculation of the same, are appropriate and in line with the applied methodology “ACM0002: Grid-connected electricity generation from renewable sources — Version 21”^{11/}. Same is also reflected in final PD version 03.1. Thus accepted.

Sub-step 2d: Sensitivity Analysis

The objective of the sensitivity analysis is to determine the likelihood of the occurrence of a scenario other than the scenario presented, in order to provide a cross-check on the robustness of the assumptions used in the development of the investment analysis

The following factor has been subjected to the sensitivity analysis:

1. Power sales revenues
2. Annual O&M costs
3. Total CAPEX
4. Power generation

The results of the sensitivity analysis show that even with a variation of +10%/-10% sensitivity parameters, the project IRR is significantly lower than the benchmark IRR. And it is evident from the results given above that the project activity remains additional even under the most favourable conditions.

The validation team confirms that, it is evident that the project is not financially viable even with the carbon benefit. PP accepts this and still has decided to go ahead with the project for environmental and social benefits as confirmed during the interview with PP. Further, PP has chosen to go ahead with the project considering a specific rate of VCUs which may increase/decrease is and is beyond the control of PP.

Percentage Variation	-10%	0%	+10%
Power sales revenues	5.28%	7.05%	8.74%
Annual O&M costs	7.32%		6.79%
Total CAPEX	7.99%		6.22%
Power generation	6.52%		7.05%

The sensitivity analysis reinforces the notion that the project's Internal Rate of Return (IRR) without carbon revenues is unlikely to meet the required benchmark of 12.54%. Due to the capped nature of power generation, an increase of 10% is deemed impossible, thus having no impact on the IRR. Exceeding such a benchmark would necessitate a significant increase in sales revenues by 34%, or a substantial decrease in investment costs by 46%. However, given the fixed Power Purchase Agreement (PPA)^{2/} and Engineering, Procurement, and Construction (EPC) terms, such scenarios are improbable. Even in the event of a 100% decrease in operating costs, the project's IRR would still not breach the benchmark, highlighting the improbability of such an outcome

Based on the submitted facilities agreement dated 23 August 2021, it is substantiated that, the date of investment decision is 23-August-2021 and considered assumptions are valid at the investment decision date.

Name of the parameter	VVB assessment
Power sales revenues	The power sales revenues are based on tariff rate/ electricity price. The Tariff rate/electricity price has been taken from the PPA^{2/} only. The VVB has also checked the PPA and found there are no changes in tariff rate/electricity price and is within threshold limit. Thus, the project activity is additional with actual Tariff rate. Verification team has reviewed and found additionality of the project could be impacted if the tariff rate increased by 34% which is not possible as the existing tariff of 0.073 EURO/kWh or 73.2 EURO MWh⁵ is fixed without escalation for the life of the project i.e. 25 years Further, VVB crosschecked considered tariff rate with tariff rate of other projects registered in other GHG mechanism i.e, GS 10869 and GS 12054, Found that tariff considered by other project are not too much deflected and within the +-10% range. Hence Accepted.
Annual O&M costs	The O&M cost has been considered from long term O&M contract order signed between <i>Quadran Burkina Faso</i> and <i>Quadran Faso services</i>. Same has been verified by VVB team and found active till now. Further, VVB team observed that O&M cost considered for the IRR calculation is the actual value paid by PP to O&M Contractor i.e.,

⁵ At Currency Conversion rate of 1 EURO = 655,957 FCFA, Same has been confirmed through Power Purchase Agreement.

	Quadran Faso Services for the first year of the operation of the project. Values has been considered as per below;																	
	<table><tr><th>Year</th><th>Yearly O&M Cost for Complete Project (in million Euro)</th><th>Per MW O&M Cost (in Million Euro)</th></tr><tr><td>Year 1</td><td>0.3503</td><td>0.0145</td></tr><tr><td>Year 2</td><td>0.3378</td><td>0.0140</td></tr><tr><td>Year 3 to Year 25</td><td>0.2640</td><td>0.0110</td></tr></table>			Year	Yearly O&M Cost for Complete Project (in million Euro)	Per MW O&M Cost (in Million Euro)	Year 1	0.3503	0.0145	Year 2	0.3378	0.0140	Year 3 to Year 25	0.2640	0.0110			
	Year	Yearly O&M Cost for Complete Project (in million Euro)	Per MW O&M Cost (in Million Euro)															
	Year 1	0.3503	0.0145															
	Year 2	0.3378	0.0140															
	Year 3 to Year 25	0.2640	0.0110															
	Above mentioned value has been verified through O&M agreements which is fixed for throughout lifetime of project activity by PP. Thus, VVB team confirm that O&M would no lesser that the value considers for IRR calculation. Thus accepted.																	
	Additionally, VVB cross checked Considered O&M cost (Under IRR calculation) with few registered projects of same region under different GHG mechanism, and observed that considered O&M cost is nearby to the applicable cost of Solar Power plant located in host country. Thus accepted.																	
	<table><tr><th>Registere d GHG Project</th><th>Country</th><th>Capaci ty (MW)</th><th>Annual O&M Cost (in million EUR) per year</th><th>Per MW O&M Cost (in Million Euro)</th></tr><tr><td>GS 10869⁶</td><td>Burkina Faso</td><td>30</td><td>0.649</td><td>0.0216</td></tr><tr><td>GS 12054⁷</td><td>Burkina Faso</td><td>25</td><td>0.565</td><td>0.0226</td></tr></table>			Registere d GHG Project	Country	Capaci ty (MW)	Annual O&M Cost (in million EUR) per year	Per MW O&M Cost (in Million Euro)	GS 10869 ⁶	Burkina Faso	30	0.649	0.0216	GS 12054 ⁷	Burkina Faso	25	0.565	0.0226
	Registere d GHG Project	Country	Capaci ty (MW)	Annual O&M Cost (in million EUR) per year	Per MW O&M Cost (in Million Euro)													
GS 10869 ⁶	Burkina Faso	30	0.649	0.0216														
GS 12054 ⁷	Burkina Faso	25	0.565	0.0226														
Based on sectoral scope expert and local knowledge, the project O&M cost considered for the proposed project activity is found to be appropriate. Also, since the O&M cost is available to VVB and IRR is still within benchmark and thus the same is acceptable.																		
VVB has compared the project Annual O&M cost with other similar projects registered in the region, ^{8, 9} and audit report submitted by third-party issued by EY & Associates/ ^{17/} and it is found the figures within the range of other projects. Thus, accepted.																		

⁶ <https://registry.goldstandard.org/projects/details/2826>

⁷ <https://registry.goldstandard.org/projects/details/2826>

⁸ <https://registry.goldstandard.org/projects/details/2826>

⁹ <https://registry.goldstandard.org/projects/details/3975>

Total CAPEX	The Total CAPEX has been considered i.e., 1 million Euro/ MW from the Facilities agreement dated 23-August- 2021. The VVB has checked the facilities agreement and found that the value considered is acceptable and is in EPC terms. Hence a decrease of 34% is an unlikely scenario to breach the threshold limit. Thus, the project activity is additional with total CAPEX being within the threshold limit VVB has compared the project Total CAPEX with other similar project registered in the region ¹⁰ and audit report submitted by third -party and it is found the figures within the range of other projects. Thus, accepted.				
	Registered GHG Project	Country	Capacity (MW)	CAPEX per MW	
	GS 10869	Burkina Faso	30	0.741 million EUR	
	GS 12054	Burkina Faso	25	1.19 million EUR	
Project Cost	The project cost has been considered from the EPC cost documents and loan facility agreement. The VVB has checked the EPC cost documents and loan facility agreement and found that the value considered is acceptable Thus, the project activity is additional with project cost being within the threshold limit				
Tax Holidays/Subsidies	VVB checked and found that this is publicly available information (tax policy/investment code of Burkina. Thus, same is acceptable.				
Power Generation	The details of the proposed project activity are given below.				
	Name of the Investor/Owner/SPVs	Site Name	Capacity (MW)	Power Generation	PLF
	Quadran Burkina Faso	Zano	24	46,244 MWh/year ¹¹	21.99 %
	Validation team assessed that the Power generation is considered as per submitted yield assessment study/ ^{9/} . Same standard has been used in the financials and the emission reduction calculation. Further, VVB crosschecked Power generation value of project activity with value of other projects registered in other GHG mechanism i.e., GS 10869 and GS 12054, Found that tariff considered by other project are not too much deflected and within the +-10% range. Hence Accepted.				
Registered GHG Project	Countr y	Capacit y (MW)	PLF	Power Generation (per year)	
GS 10869	Burkin a Faso	30	18.64 %	49 GWh or 49,000 MWh	

¹⁰ <https://registry.goldstandard.org/projects/details/2826>

¹¹ Power generation is based on yield assessment study (which is before applying degradation factor)

	GS 12054	Burkina Faso	25	16.89 %	37 GWh or 37,000 MWh
	Assessment team noted that increment of 10% in the power generation is unlikely scenario. Thus, the project activity is additional with the power generation being within the threshold limit.				

Taking the above assessment into consideration assessment team concludes that the IRR is robust to the sensitive parameters, and the results of sensitivity analysis show that even with a variation of +10% & -10% in project cost, O&M cost, Tariff Rate, PV Production, The IRR is significantly lower than the benchmark. Therefore, it is evident from the results given above; the project remains additional even under the most favorable conditions. Assessment team also confirmed the breaching values for individual parameters and thus confirms that the project is still additional.

Step 3: Barriers analysis;

The project proponent has the option to utilize either the investment analysis or barrier analysis step. Since the project proponent has already implemented the investment analysis, there is no necessity to elaborate on the barriers analysis.

Step 4: Common practice analysis

As per the “Tool for demonstration and assessment of additionality” (Version 07.0.0), Step 2 Investment Analysis shall be complemented with an analysis of the extent to which the project type (e.g., technology or practice) has already diffused in the relevant sector and region within the country. This test is a credibility check to complement the investment analysis. The Methodological Tool followed in this Analysis is Methodological Tool 24: Common Practice, Version 3.1./^{11/}

Common Practice Analysis:

Step (1): Calculate applicable capacity or output range as +/-50% of the total design capacity or output of the proposed project activity.

Range	Capacity (incl. phase 2)	Unit
+50%	36	MW
Project activity capacity	24	MW
-50%	12	MW

Step (2): Identify similar projects (both CDM and non-CDM) which fulfil all of the following conditions:

- the projects are located in the applicable geographical area
- the projects apply the same measure as the proposed project activity
- the projects use the same energy source/fuel and feedstock as the project activity, if a technology switch measure is implemented by the project activity

- the projects implemented produce goods or services with comparable quality, properties and applications areas (e.g., clinker) as the project plant
- the capacity or output of the projects is within the applicable capacity or output range calculated in
- the projects started commercial operation before the project documents have been published for global stakeholder consultation or before the start date of the project activity, whichever is earlier for the proposed project activity.

Analysis of Step 2

Identification of the similar projects (CDM and non-CDM) is carried out as per sub-steps of Step (2) as follows:

- as the project is located in Burkina Faso, projects in the geographical area of Burkina Faso have been chosen for the analysis

The validation team noted that as the project is located in Burkina Faso, therefore, projects in the geographical area of Burkina Faso have been chosen for analysis. Each country has different policies regarding renewable energy, hence considered as geographical region for common practice analysis. The distinction from choosing the Nation to entire geographical boundary is already explained above in the report and thus the applied geographical area is acceptable to the validation team.

- the project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying the same measure (b) are candidates for similar projects

The validation team noted that the project activity is a green-field solar power project and uses measure (b) “Switch of technology with or without change of energy source including energy efficiency improvement as well as use of renewable energies”. Therefore, projects applying same measure (b) are candidates for similar projects.

- the energy source used by the project activity is solar. Hence, only solar energy projects have been considered for the analysis

The validation team confirms during the remote audit that the energy source used by the project activity is solar. Hence, only solar energy projects have been considered for analysis.

- the project activity produces electricity; therefore, all power plants that produce electricity are candidates for similar projects

The validation team confirms during the remote audit that the project activity produces electricity; therefore, all power plants that produce electricity are candidates for similar projects.

- the capacity range of the projects is within the applicable capacity range from 12 MW to 36 MW

Since the project activity is 24 MW, the output range of +/- 50% has been considered as 36 MW (Higher range for comparison) to 12 MW (Lower range for Comparison) which is assessed to be correct by the validation team.

- as per the methodological tool the projects that have started commercial operation before the project documents have been published for global stakeholder consultation or before the start date of the project activity, are to be selected for consideration

The start date of the project activity is 24-october-2023. The projects which have started commercial operation till VCS PD preparation date have been considered for analysis.

The findings of the analysis of Step 2

In the host country of Burkina Faso, the similar power plants operating before the start date of the project and belonging to the 12-36 MW output range are :

Power plant	Year	Installed capacity	Technology	Project ID
Zina Solar PV power plant	2014	22.78 MW	Solar	CDM 10151
Zagtouli PV	2017	33 MW	Solar	CDM (CPA 10320-P1-0003-CP1)
Off-grid Solar PV project at IAMGOLD Essakane SA Gold Mine	2018	15 MW	Solar off grid	CDM 10398
Nagréongo	2022	30MW	Solar	GS 10869

Step (3): Within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing project verification.

$N_{all} = 0$

Step (4): Within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity.

$N_{diff} = 0$

As per the requirement of step 3, CDM project activities, which have got registered or are under validation, have been excluded in this step. The source of check for the same is cdm.unfccc.int

Step (5): Calculate factor $F = 1 - N_{diff}/N_{all}$ representing the share of similar projects (penetration rate of the measure/technology) using a measure/technology similar to the measure/technology used in the project activity that delivers the same output or capacity as the project activity.

$F = 1 - N_{diff}/N_{all}$

$F = 1 - \infty$

$= \infty$

Thus, against the requirements set in the Methodological Tool 24: Common Practice version 3.1/^{11/}, being $F = \infty$, and $N_{all} - N_{diff} = 0$, it can be concluded that the project activity is not a “common practice” within the applicable geographical area.

The above discussions show that solar power development is not a common practice and the project activity is not financially attractive; hence the project activity is additional. Thus, common practice analysis is acceptable to the validation team.

Conclusion:

The validation team has checked the O&M cost as per the contract and found it is within the threshold limit, considering the project's O&M cost and its escalation as per the O&M contract for the proposed project activity, which is found to be appropriate for solar power projects. Additionally, since the O&M cost is available to VVB and IRR is still within benchmark, the same is acceptable. It's noted that the O&M cost is not a major factor that could impact the project's IRR.

The validation team assessed that the annual generation considered and derived from acceptable yield assessment study/^{9/} for the project activity is within the range of sensitivity analysis and has been defined correctly in a VCS PD/^{7/}, which was checked and confirmed to be in line with the guidance provided in the investment analysis tool.

Hence, it can be concluded by the validation team that the project activity is not a common practice and thus is considered additional.

3.3.6 Quantification of GHG Emission Reductions and Removals

The validation team checked the baseline, project and leakage calculation and confirms that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detailed analysis is as below:

According to the applied methodology, ACM0002 Large Scale Consolidated Methodology: Grid-connected Electricity Generation from Renewable Sources, Version 21.0/^{11/}, since the project activity is the installation of Greenfield Renewable Energy Power Plant, the baseline scenario is the electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in “TOOL07: Tool to calculate the emission factor for an electricity system” version 07.0./^{11/}

The scope and applicability of Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool, version 01.0, /^{11/} covers the grid emission factors for the electricity system of the WAPP (West African Power Pool).

This Standardized Baseline can be used under the following conditions.

Baseline Procedure	Justification
The project activity is implemented in any one of the WAPP member countries i.e., Benin, Burkina Faso, Côte d'Ivoire, Ghana, Mali, Niger, Nigeria, Senegal and Togo and is connected to the WAPP	The project activity is in Burkina Faso and is connected to the WAPP.
The CDM approved methodology that is applied to the project activity requires the determination of CO ₂ emission factor(s) through the application of the grid tool	ACM0002 requires the determination of CO ₂ emission factor(s) through the application of the grid tool "TOOL07: Tool to calculate the emission factor for an electricity system version 07.0
The project activity uses the ex-ante options for both the operating margin and build margin grid emissions factors, as described in the grid tool, and therefore no monitoring or recalculation of the emission factor during the crediting period is required	The project activity uses the ex-ante options for both the operating margin and build margin grid emission factors, as described in the grid tool hence No-monitoring or recalculation of the emission factor is not required during the crediting period.

According to the ACM0002, Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel-fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above the baseline levels would have been generated by existing grid-connected power plants and the addition of new grid- connected power plants.

According to para 4 of the Standardized baseline: Grid emission factor for the West African Power Pool, version 01.0, the latest approved standardized baseline valid value of the CO₂ emission factor is 0.573 tCO₂/MWh applied for the project electricity system since Togo is listed in the WAPP member countries.

The baseline emissions are to be calculated as follows:

$$B_{ey} = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

$$B_{ey} = \text{Baseline emissions in year } y \text{ (tCO}_2\text{/yr)}$$

$$EG_{PJ,y} = \text{Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year } y \text{ (MWh/yr)}$$

$E_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of “TOOL07: Tool to calculate the emission factor for an electricity system” version 07.0 (tCO₂/MWh) and as per the Standardized baseline: Grid emission factor for the West African Power Pool, version 01.0, the Combined margin CO₂ emission factor for the WAPP power system for solar power generation project activities applicable for first crediting period in tCO₂/MWh

Calculation of $EG_{PJ,y}$

In case of Greenfield power plant development, the calculation of the electricity system’s combined margin (CM) shall be based on the following equation, according to the methodology:

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)

$EG_{facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Calculation of $EF_{grid,CM,y}$

As stated before, the “TOOL07: Tool to calculate the emission factor for an electricity system” allows referring to the Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool version 01.0. /11/

The ASB0034-2021 Standardized Baseline states that for wind and solar power generation project activities, the $EF_{grid,CM,y}$ to be used is 0.573 tCO₂/MWh.

Project Emission

The project activity involves renewable power generation. Therefore, $PE_y = 0$. This is found acceptable.

Leakage Emission

No leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g., extraction, processing, transport etc.) are not calculated being negligible and this is found acceptable by the VVB.

Net GHG Emission Reductions and Removals

Baseline Emissions:

Baseline Emissions are calculated as follows:

$$BE_y = EG_{pj,y} * EF_{grid,CM,y}$$

Where:

BE_y = Baseline emissions in year y (tCO₂ /year)

$EG_{pj,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)

$EF_{grid,CM,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y. The value is assumed based on the Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool calculated using version 01.0 of the “Tool to calculate the emission factor for an electricity system” (tCO₂/MWh).

Parameters determined ex-ante:

Data / Parameter	EF grid, CM, y
Data unit	tCO ₂ /MWh
Description	Combined Margin CO ₂ grid emission factor in the year y
Source of data	Standardized Baseline ASB0034-2021
Value applied	0.573
Justification of choice of data or description of measurement methods and procedures applied	The availability of a Standardized Baseline for the Grid Emission Factor for West African Power Pool version 01.0
Purpose of Data	Calculation of baseline emissions
Comments	This parameter is fixed ex-ante for the entire crediting period

Data and Parameters Monitored:

Data / Parameter	EG facility
Data unit	MWh/year
Description	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Source of data	Measured directly with electricity meter(s) at project site.
Description of measurement methods	Electricity outputs will be electronically stored and invoiced monthly to Sonabel by Quadran Burkina Faso, after reconciliation

and procedures to be applied	with internal meter which readings are recorded on a record sheet by the Technical/Engineering/ Maintenance Department under the Plant Manager's authority
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	44,659
Monitoring equipment	Bi-directional electronic electricity meters measuring the net electrical energy delivered to Sonabel with the following specifications: Make: Prometer Model: Prometer100 SN: WP050493 Accuracy class: 0.2S (±0. 2%) Backup meter : Landis+Gyr E650 SN: 59383221 Accuracy class: 0.2S
QA/QC procedures to be applied	Cross check of measurement results with invoiced electricity. Calibration of the meters will be carried out after each deviation of more than +- 2% between the 2 meters as per manufacturer & PPA specifications. A verification of the meters shall be done by a certified third party every three years
Purpose of data	Calculation of baseline emissions
Calculation method	Electronic recording
Comments	-

Based on the meter readings and the monthly recordings provided by the PP to the assessment team the quantity of net electricity generated and supplied to the grid is 44,659 MWh/Year by the solar plant.

The total estimated quantity of net electricity supplied to the grid:

$$EG_{pj,y} = 44,659 \text{ MWh/year}$$

The Grid Emission Factor is:

$$EF_{grid,CM,y} = 0.573 \text{ tCO}_2/\text{MWh}$$

The calculation of the baseline emissions of the project activity is:

$$BE_y = 44,659 \text{ MWh/yr} * 0.573 \text{ tCO}_2/\text{MWh}$$

$$= 25,589 \text{ tCO}_2/\text{yr.}$$

Leakage:

Leakage emissions are not applicable.

Project Emissions:

Project Emissions are as shown below:

$$PE_y = 0 \text{ tCO}_2/\text{yr}$$

Emission Reductions:

$$ER_y = BE_y - PE_y$$

$$ER_y = 25,589 \text{ tCO}_2/\text{yr} - 0 \text{ tCO}_2/\text{yr}$$

$$= 25,589 \text{ tCO}_2/\text{yr}$$

Year	Estimated baseline emissions or removals (tCO _{2e})	Estimated project emissions or removals (tCO _{2e})	Estimated leakage emissions (tCO _{2e})	Estimated net GHG emission reductions or removals (tCO _{2e})
24-October-2023 to 31-December-2023	4,936	0	0	4,936
1-January-2024 to 31-December-2024	25,967	0	0	25,967
1-January-2025 to 31-December-2025	25,825	0	0	25,825
1-January-2026 to 31-December-2026	25,682	0	0	25,682
1-January-2027 to 31-December-2027	25,541	0	0	25,541
1-January-2028 to 31-December-2028	25,401	0	0	25,401
1-January-2029 to 31-December-2029	25,261	0	0	25,261
01-January 2030 to 23-October -2030	20,511	0	0	20,511
Total	179,124	0	0	179,124

(7 Years)				
-----------	--	--	--	--

3.3.7 Methodology Deviations

The validation team confirms that no methodology deviation is applied to the project activity.

3.3.8 Monitoring Plan

The proposed Monitoring Plan describes the Management System and Procedures to be developed and implemented by the project owner for this project in order to ensure consistency with the methodology ACM0002: Grid-connected electricity generation from renewable sources - -- Version 21.0 which is applied in the context of the project activity.

The developed monitoring plan contains the following sections:

1. Monitoring team and training
2. Quantity management procedures

All data required for the calculation of emission reductions associated with the project activity will be monitored in accordance with the approved methodology ACM0002: Grid-connected electricity generation from renewable sources --- Version 21.0

Monitoring team and Training

A dedicated team, well-versed in VCS requirements, will handle data collection, consolidation, and results analysis. This team will operate independently, without hierarchical or dependence links to any involved entities. Their tasks include measuring net electricity supplied to the grid and ensuring the proper operation and maintenance of measuring equipment. This independence is crucial to maintaining the integrity of the work conducted.

Quality management procedures

Establishing a real-time supervision and control system to transmit monitoring data off-site ensures that operators can securely access and retrieve all project data, even in the event of incidents.

Meter measurements conducted by SONABEL main/back-up must not deviate beyond $\pm 2\%$. Any dysfunctional meters exceeding this limit must be promptly identified by SONABEL and Quadran Burkina Faso and adjusted or replaced within 48 hours following manufacturer guidelines. For deviations exceeding $\pm 1\%$, each meter will undergo testing and calibration certified by a third party. During non-calibrated periods, non-conformance data will be systematically discounted by the corresponding error margin.

The validation team checked the monitoring practice followed on-site during the remote audit and confirmed that it is in line with the defined monitoring plan as per VCS PD, The VVB is of the

opinion that, the project activity is being monitored adhering to the monitoring plan to the requirements of the applied methodology and referred tools.

3.4 Non-Permanence Risk Analysis

Not applicable for the project activity.

4 VALIDATION OPINION

LGAI Technological Center, S.A. (Applus+ Certification) has been engaged by “AERA Group” to perform the validation of the project activity “Zano Solar Power”.

The management of the Project Proponent/owner is responsible for the preparation of the GHG emissions data and the reported/estimated GHG emissions reductions on the basis set out within the project’s Monitoring Plan in the VCS PD/⁷and the approved methodology ACM0002: Grid-connected electricity generation from renewable sources — Version 21/¹¹.

Our Validation approach was based on the requirements as defined by the VCS board. Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these.

The Validation Team determined that the claims made for this VCS Project are according to the VCS regulatory requirements and are suitable to occur during the operation of the Project, hence the claims are reasonable, including the assumptions, limitations and methods established by the PP supporting these claims.

The validation team confirms that:

- The project is likely to achieve the estimated level of Emission Reductions and any other claims as stated by the PP.
- Actual results may vary since the estimates are based on assumptions that are subject to change.
- The project description complies with the requirements and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS Program Guide version 4.4 and VCS Standard version 4.5.
- The project’s baseline and additionality are assessed against “ACM0002: Grid-connected electricity generation from renewable sources — Version 21 and Tool for the demonstration and assessment of additionality version 07.0,
- The project’s monitoring plan is assessed against “ACM0002: Grid-connected electricity generation from renewable sources — Version 21

- The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews with Project Owner have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence for positive validation opinion as per the requirement of VCS.

- The project is expected to generate 25,589 tCO₂e during the length of entire crediting period (24- October-2023 to 23-October-2030).

Validated GHG emission reductions and removals period for the 1st crediting period

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 2023 (From 24-October-2023 to 31-December-2023)	4,936
Year 2024 (1-January-2024 to 31-December-2024)	25,967
Year 2025 (1-January-2025 to 31-December-2025)	25,825
Year 2026 (1-January-2026 to 31-December-2026)	25,682
Year 2027 (1-January-2027 to 31-December-2027)	25,541
Year 2028 (1-January-2028 to 31-December-2028)	25,401
Year 2029 (1-January-2029 to 31-December-2029)	25,261
Year 2030 (From 1-January-2030 to 23- October-2030)	20,511
Total estimated ERs	179,124
Total number of crediting years	7 Years
Average annual ERs	25,589

APPENDIX 1: DOCUMENTS REVIEW UNDER VALIDATION

No.	Author	Title	References to the document	Provider
1.	N/A	Statistique production report	Commercial operation date (As grid connection date 24-October -2023)	Project Proponent
2.	N/A	Ownership Documents	Power Purchase Agreement, dated 20-November-2019 (between LA SOCIETE NATIONALE D'ELECTRIC DU BURKINA (SONABEL) and QUADRAN BURKINA FASO	Project Proponent
3.	N/A	Long term O&M agreement	1. Monitoring manual of the project site 2. O&M agreement of Quadran Burkina Faso (the owner) and Quadran Faso Services (the operator) ¹²	Project Proponent
4.	N/A	Facilities Agreement	Dated 23-August-2021	Project Proponent
5.	Applus	Contract of the PP with the VVB	Dated 21- March- 2023	Project Proponent
6.	Longi Solar	Technical specifications	Technical specifications from manufacturers	Project Proponent
7.	PP	VCS PD version 04	13-September-2024	Project Proponent
8.	PP	Emission Calculation sheet- version 03	05- January -2024	Project Proponent
9.	ARTELIA	Yield assessment study (initial report)	Dated 29-May-2020	Project Proponent

¹² For 2 years and may be extend up to 7 years

No.	Author	Title	References to the document	Provider
10.	NA	Reference link is provided.	https://solarquarter.com/2024/01/12/burkina-faso-unveils-zano-photovoltaic-solar-power-plant-boosting-clean-energy-goals/ https://www.power-technology.com/data-insights/power-plant-profile-zano-solar-pv-park-burkina-faso/#:~:text=Zano%20Solar%20PV%20Park%20is,developed%20in%20a%20single%20phase	Independent Search
11.	UNFCCC VERRA	Tools/ guidelines used in the project activity	UNFCCC CDM web site • UNFCCC Methodology: ACM0002: Grid-connected electricity generation from renewable sources --- Version 21 • Tool: Tool for the demonstration and assessment of additionality version 07.0 • VCS guideline version 4.4 • VCS Project standard 4.5 • Standardized Baseline ASB0034-2021: Grid emission factor for the West African Power Pool calculated using version 01.0 • TOOL01: Tool for the demonstration and assessment of additionality version 07.0 • Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 3.0 • Tool 07: Tool to calculate the emission factor for an electricity system, version 7.0 • Tool 24: Common practice, version 03.1 • Tool 27: Investment analysis, version 12.0	UNFCCC
12.	PP	VCS Declaration	Declaration from PP for Participation under Other GHG Programs	Project Proponent
13.	PP	LSC	Local stakeholders' consultation supporting; Notices, MoM's & photographs etc. dated 21-February-2020	Project Proponent

No.	Author	Title	References to the document	Provider
14.	VERRA	VCS site exemption	Communication letter dated 31-July-2023	Project Proponent
15.	PP	VCS Declaration	No ODA declaration, Dated 30-May-2023	Project Proponent
16.	PP	ESIA	ESIA report, Dated 31-January-2019	Project Proponent
17.	EY & Associate	Audit Certificate	Dated 17/12/2021	Project Proponent
18.	PP	IRR Sheet Both Confidential and Non-Confidential	Dated 11/08/2021	Project Proponent

APPENDIX 2: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)

CL ID	01	Section no.	3.3.5	Date: 17-january -2024
Description of CL				
PP is requested to clarify the date of the investment decision. Additionally, PP is to mention the reference document for verifying the date of the investment decision and provide the document as well.				
Project Proponent response				Date: 18-January-2024
<i>The investment decision in 23 August 2021, as the date of signature of the facilities agreement.</i>				
Documentation provided by Project Proponent				
Document already provided, but resubmitted "20230821 Extrait facility agreement QBF-FMO-PPCO"				
VVB assessment				Date: 25-January-2024
The assessment team checked and confirmed that PP has submitted the facilities agreement dated 23 August 2023, which is also considered as the investment decision date. This finding was deemed acceptable to the VVB. Hence, this CL is closed.				
CL 01 is closed				

CAR ID	01	Section no.	3.1	Date: 04-Aug-2023
Description of CAR				
During review of monitoring report following inconsistencies observed:				
- The title of the project activity given in the cover page is found incorrect. Thus, corrective action sought. - Section 1.1 is not inline with the VCS PD Template guidelines V 4.2. Thus, corrective action sought. - Date format is found inconsistent throughout the VCS PD. Thus, corrective action sought. - Audit history table is found missing in Section 1.1 of the VCS PD. Thus, corrective action sought. - PP is requested to submit PPA in order to confirm the legal right to control and operate the project activity of Zano. Kindly submit. - PP is requested to submit technical specification details of the project activity to the assessment team. Kindly submit. - PP requested to submit evidences to cross-check the start date of the project activity. Kindly submit. - There are three project proponent involved in this project (QAIR, Quadran Burkina Faso and Dedougou Solaire). PP shall clarify out of these three, which company will have legal ownership on the VERs generated and which PP will be the focal point of communication with the registry. - PP shall provide copy ESIA report and Biodiversity management plan. - During remote audit it was observed that the project sites located on an agricultural land and people were relocated from the project area. PP shall provide all the necessary permits/ licenses/ approvals required for implementation of the project and ensure Resettlement and Rehabilitation of PAPs (Project Affected People) were done in accordance with the host country regulations.				
Project Proponent response				Date: 24- Aug-2023

8. legal ownership clarified in section 1.5. AERA Group is the project proponent, Qair Burkina Faso and Dedougou are the SPV's of the plants, Qair is the main sponsor. 9. ESIA and biodiversity management plan now provided 10. compensation and reinstallation plans now provided.	
Documentation provided by Project Proponent	
1) PD version 3 2) ESIA and biodiversity management plan. 3) Resettlement and rehabilitation of PAPS reports	
VVB assessment	Date: 28- Aug-2023
The validation team has reviewed the PD version 3.0 dated 24/08/2023 points as mentioned below. 1. The title of the project is corrected. Same has been verified and accepted, 2. Section 1.1 has been corrected as per the VCS PD Template guidelines V4.2. Hence it is accepted. 3. Date format is corrected by PP, verified & found consistent throughout the VCS PD. 4. Audit table has been added by the pp 5. PP has submitted PPA 5.4 Annexe 4 - Extrait Contrat PPA Zano to confirm the legal right to control and operate the project activity of Zano, same has been verified and accepted. 6. The PP has submitted the technical specifications 202201 Datasheet PV Module, 202201 SG250HX Datasheet Inverter and Prometer 100 _Data Sheet of the project activity of Zano. 7. As per the VCS PD the project start date is 1 st September 2023 as the expected date of commissioning of the project. PP has to submit the document of start date of the project 8. As per VCS PD the validation team has reviewed the documents of PPA of Zano submitted by the PP. The legal right to control and operate the project activity of Zano belongs to Quadran Burkina Faso as per PPA 20 November 2019. The legal right to trade the emission reductions belongs to AERA Group as per ERPA dated 8 November 2021 for Zano. 9. The validation team has reviewed the documents ESIA of Zano dated 31/01/2019 and Biodiversity management plan of Zano dated 28/08/2021 10. The validation team has reviewed the PP has submitted the document compensation and reinstallation plan Zano (Project no.: 0514729) dated 16/06/2020. Same has been accepted.	
VVB assessment	Date: 17-January-2024
7. As per the VCS PD the project start date is 24 th October 2023 as the expected date of commissioning of the project. Since the expected date has been passed, thus, PP to revise the date. Or, if it is commissioned, PP to provide the supporting document (Commissioning certificate) of the project activity. Hence CAR-01 is open	
Project Proponent response	Date: 23-January-2024
7. At the moment PP does not have a COD certificate, but has the statistic report production from the site that shows a start of production on 24/10/2023.	
Documentation provided by Project Proponent	
VCS PD	
VVB assessment	Date : 25-Jaunary - 2024
On the basis of Statistic report, Assessment team has verified that the PP has started production since 24/10/2023 which has been considered as the start date of the project activity. Hence, this CAR is closed. CAR 01 is closed	

CAR ID	02	Section no.	3.2.2	Date: 24-May-2023
Description of CAR				
PP is requested to provide complete details of LSC & submit records of LSC meeting & ongoing local stakeholder consultation including grievance register etc. Kindly submit.				

Project Proponent response	Date: 24- Aug-2023
<i>Please see documentation provided.</i>	
Documentation provided by Project Proponent	
<i>0514729 ERM Insuco Quadran PEPP Zano_FINAL 16062020</i>	
VVB assessment	Date: 28- Aug-2023
The validation team has reviewed the detailed report of Local stakeholders' engagement plan of Zano dated 21/02/2020. The validation team concludes that the plan complies with the VCS requirement. Hence it is accepted. Hence CAR-02 is closed.	

CAR ID	03	Section no.	3.3	Date: 04-Aug-2023
Description of CAR				
1. PP is requested to submit supporting for additionality under section 3.5 of VCS PD. PP shall provide supporting documents/ references for all the input parameters considered for investment analysis available at the time of decision making. 2. PP requested to revise the applied methodology ACM0002 Version under Section 4.4. Corrective action sought. 3. Under Section 1.16.2, 1.18, 2.2 the details are not correctly filled. Hence, found not inline with VCS PD Template guidelines V 4.2. Thus, corrective action sought.				
Project Proponent response				Date: 24- Aug-2023
<i>1. Please see below documents provided</i>				
Documentation provided by Project Proponent				
<i>PPP and PPA for both SPV's</i> <i>Price of electricity</i> <i>Facility agreement</i> <i>Short term OM Zano</i>				
VVB assessment				Date: 28- Aug-2023
1. PP has not submitted enough supporting document for additionality under section 3.5 of VCS PD and also not clearly defined Assumption with source documents as reference are not clearly defined in IRR sheet. In facility agreement the values are not mentioned with units. 2. The validation team has verified that the PP has corrected the applied methodology ACM0002 latest version 21.0. Hence is accepted. 3. The PP has not updated the details in sections 1.16.2, 1.18,2.2 . Hence it is not in line with VCS PD. Hence CAR-03 is open.				
Project Proponent response				Date: 30-Oct-2023
1. In the facility agreement, the head column states EURO*1000 which is the unit. Other document listed in IRR link figure / document excel sheet. Documents sent to VVB. 3. Sections 1.16.2 and 1.18.2.2 updated.				
VVB assessment				Date: 17-January-2024
1. PP response is accepted. However, PP to insert the input parameters and units and respective sources of the parameters with weblink/supporting documents. Hence, this part of CAR is open. 3. Revised PDD was checked and confirmed that PP has revised the respective sections which is found to be correct. Hence, this part of CAR is closed. Hence, CAR 03 is open				
Project Proponent response				Date: : 23- January-2024

Input parameters and units, and respective sources of the parameters inserted as two new tabs in the IRR sheet. And updated in PDD table 8	
Documentation provided by Project Proponent	
IRR sheet	
VVB assessment	Date: : 25-January -2024
It was checked and confirmed that PP has submitted revised IRR sheet. Also, in the revised PDD, PP has inserted input parameters. Hence, this CAR is closed. CAR 03 is closed	

CAR ID	04	Section no.	3.2	Date: 04-Aug-2023
Description of CAR				
PP is requested to submit an undertaking for no double accounting for project activity for participation in other GHG program other than VCS as well as host country/international Renewable Energy Certificate (REC) mechanisms. Kindly submit.				
Project Proponent response				Date: 24- Aug-2023
Documentation provided by Project Proponent				
20230530 VCS validation of Zano Solar Power BF Declaration on double counting				
VVB assessment				Date: 28- Aug-2023
The validation team has assessed the declaration on No double accounting made by the PP vide reference no. 3005-2/QBF/DGA/DP/MAY/2023 dated 30/05/2023 is found in line with VCS requirement. Hence it is accepted. Hence CAR-04 is closed				

CAR ID	05	Section no.		Date: 04-Aug-2023
Description of CAR				
PP requested to submit estimated Emission Reduction sheet. Kindly submit.				
Project Proponent response				Date: 24- Aug-2023
See documentation provided				
Documentation provided by Project Proponent				
221117 Er Ex ante calc-ZANOSO				
VVB assessment				Date: 28-Aug-2023
The PP has submitted estimated Emission reduction sheet of the project activity. Hence CAR-05 is closed				

CAR ID	06	Section no.	3.3.8	Date: 04-Aug-2023
Description of CAR				
VCS PD found missing complete information of metering equipment and calibration frequency along with monitoring arrangements at site inline with requirements of applied methodology. Thus, PP requested to revise and resubmit VCS PD.				
Project Proponent response				Date: 24- Aug-2023

S/N now added to section 5.2.	
Documentation provided by Project Proponent	
VCS-PD	
VVB assessment	Date: 28- Aug-2023
The validation team has reviewed the VCS PD v3.0 section 5 / 5.2 in that information on metering details like Serial no. , Accuracy and make etc. of Main meter, check meter and Standby meter details were not mentioned. However, the calibration frequency along with monitoring arrangements at site in line with requirements of applied methodology is mentioned. Hence this CAR-06 is open	
Project Proponent response	Date:27-Nov-2023
S/N now added to section 5.2.	
VVB assessment	Date: 30-Nov-2023
The validation team has reviewed the VCS PD v3.0 section 5.2 in that information on metering details like Serial no., Accuracy and make etc. of only Main meter details were mentioned. However, PP to provide the supporting document as well. Hence this CAR-06 is open	
Project Proponent response	Date :23-January-2024
Technical specification now provided	
Documentation provided by Project Proponent	
LandisGyr E650 Brochure Prometer100_wall	
VVB assessment	Date :25-January-2024
Submitted document was checked and found consistent with the mentioned technical specification. Hence, this part of CAR is closed. CAR 06 is closed	

CAR ID	07	Section no.	3.3.5	Date: 17-January-2024
Description of CAR				
1. In the IRR sheet, PP to provide the calculation of benchmark and for sensitivity analysis also. Correction is required. 2. In the IRR sheet PP to mention source documents for various assumptions are in English. Correction is sought.				
Project Proponent response				Date: 23 January 2024
Input parameters and units, and respective sources of the parameters inserted as two new tabs in the IRR sheet which has the sensitivity analysis and benchmark calculation.				
Documentation provided by Project Proponent				
VVB assessment				Date: 25-January-2024
1. Assessment team has checked and confirmed that in the revised IRR sheet, PP has inserted the calculation of benchmark and sensitivity analysis. Hence, this part of CAR is closed. 2. Revised IRR sheet was checked and confirmed that PP has been mentioned source documents for various assumptions are in English. Hence, this part of CAR is closed. CAR 07 is closed				

FAR ID	01	Section no.	3.1	Date: 03-February-2024
Description of FAR				

Verifying VVB to check the commissioning certificate to verify the commissioning date of the project activity at the time of verification.	
Project Proponent response	Date: DD-Month-YYYY
Documentation provided by Project Proponent	
VVB assessment	Date: DD-Month-YYYY

APPENDIX 3: COMPETENCE OF TEAM MEMBER AND TECHNICAL REVIEWER

Mr. Pankaj Kumar has done M. Sc in Environment Management from Forest Research Institute, Dehradun and B. Sc. (Hons.) in Environment & Water Management from Magadh University, Bihar, India. He has also done Post Graduate Diploma in Environmental Law from NLSIU, Bangalore.

He has more than 12 years of working experience in GHG Assessments and has participated during his career in Agencies and DOEs like MITCON, Agrinergy, Carbon Check and is empaneled with Applus+ Certification since 2015 for the performance of CDM/VCS/GS project assessments.

He has extensive experience in the Renewable, Waste Management and Energy Demand Scopes of UNFCCC CDM and has done more than 100 Validations and Verifications of PAs and PoAs as Lead Auditor, Technical Expert and Technical Reviewer, mainly in Asia, Africa, USA, Asia Pacific and Americas under CDM, Verified Carbon Standard, Gold Standard & Social Carbon Standard, Brazil.

He is an experienced, qualified and result oriented Environment and climate change professional having 16 yrs. of relevant experience in Climate Change (Mitigation & Adaptation), Environmental Due Diligence, Disaster Risk Reduction, Climate finance, adaptation planning, capacity building, validation and verification of GHG project. He can also provide technical support for environmental investigative, remedial projects involving air, water and soil, Waste management, EIA, Environmental Compliance, ISO 14001, OHSAS 18001, GHG accounting (ISO 14064) and Carbon foot printing.

Mr. Srikanth Meesa is a climate change professional with more than 17 years of experience in the fields of climate change, GHG auditing, carbon footprint assessments, water and energy audits, scope-3 emissions, net zero strategies, sustainability & ESG sectors and water and wastewater treatment sectors. He has completed his Master's in Environmental Engineering & Management from I.I.T Delhi and his Bachelor's in Civil Engineering from Osmania University. Furthermore, he has extensive experience i.e., more than 14 years in the auditing of CDM, VERRA, GS, JCM projects of various sectoral scopes across the world. He has worked with reputed certification bodies such as TUV NORD, DNVGL, TUV SUD, and LR. He also has consulting experience on sustainability reports & road-maps, stakeholder consultation, and LCA studies while working at Thinkstep (Now called Sphera). Currently, he is associated with Global Green Solutionz and empaneled with Applus+ Certification to carry out GHG audits in the aforementioned schemes. Mr. Srikanth Meesa is based at Hyderabad, India.

APPENDIX 4: ABBREVIATIONS

Abbreviations	Full texts
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CL	Clarification request
CM	Combined Margin
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
DNA	Designated National Authority
DR	Document Review
EF	Emission Factor
EIA	Environmental Impact Assessment
ER	Emission Reductions
FAR	Forward Action Request
GHG	Greenhouse gas(es)
GWP	Global Warming potential
SONABEL	LA SOCIETE NATIONALE D'ELECTRICITE DU BURKINA
VVB	Validation/Verification Body

APPENDIX 5: STAKEHOLDER CONSULTATION

A Local Stakeholder Consultation process has been conducted in accordance with the VERRA- Verified Carbon Standard (VCS) requirements, towards the successful registration of emission reduction projects/programs aiming to generate carbon credits for the project activity.

The Local Stakeholder Consultation process took place as follows:

The Local Stakeholder Consultation process was held on 21-February-2020 at Zano and attended by Prefect (Village Chief), local persons including local villagers, local vendors, and technology suppliers. All the stakeholders were invited through an invitation letter dated 07-February-2020 to attend the stakeholders meeting.

The stakeholders identified by the Project Proponent were local villagers who are the major population of the area, local communities, and Prefect (Village Chief), EPC Contractor, project proponent representatives, O&M Team and other people involved in the project. Validation team verified the list of participants who attended the stakeholder meeting and feedback questionnaire and confirms the stakeholders identified are relevant.

Validation team verified the list of participants who attended the stakeholder meeting and feedback questionnaire and confirms the stakeholders identified are relevant.

The validation team also verified the minutes of meeting to note that no negative comments were received and the same was cross-checked with the information obtained during follow-up interviews with the stakeholders.

Thus, VVB concludes that, the stakeholder's consultation was conducted in accordance with the VERRA- Verified Carbon Standard (VCS) requirements, towards the successful registration of emission reduction projects/programs aiming to generate carbon credits for the project activity.